WORKSHOP

# What predicts lexical decision speed?

Today's question is a psycholinguistic one, not just a statistical one

We will test whether classic lexical variables predict how quickly words are recognised

We will use regression as a tool for answering a theory-driven question

By the end, you should be able to turn output into a short analysis-style interpretation

WHY IT MATTERS

# Why this workshop matters for the block and the exam

**Lecture 1** asked how lexical variables shape word recognition

**Lecture 2** asked how words become integrated into the lexicon over time

Today's dataset focuses on **established words**, so the main emphasis is Lecture 1

The bridge to Lecture 2 is that these predictors reflect the structure of an **integrated lexical system**

This workshop uses the same sequence of reasoning as the **exam analysis question**, without functioning as a mock exam

RETRIEVAL

# Before we open the data

1. What does lexical decision measure?
2. Why does word frequency usually affect RT?
3. What does AoA capture that frequency may not fully capture?
4. What is lexical competition?
5. What would count as evidence that a new word has become integrated?

PREDICTIONS FIRST

# What should happen?

- Predict the direction of each relation with RT
- Higher **log frequency** should predict...?
- Higher **AoA** should predict...?
- Greater **length** should predict...?
- Higher **OLD20** should predict...?
- Which predictors do you expect to survive when they are entered together?

DATASET ORIENTATION

# What each column means

**length**
number of letters

**log_frequency**
how common the word is in the language (**higher = more frequent**)

**AoA**
estimated age at which the word is learned (**higher = learned later**)

**OLD20**
a measure of orthographic neighbourhood similarity

**mean_RT_ms**
average lexical decision response time

Also in the file: **word**, **concreteness**, **morph_family_size**, **mean_accuracy**

Not every variable enters today's core model; the live analysis focuses on **length, log_frequency, AoA, and OLD20**

ANALYTIC HYGIENE

# Sanity checks before modelling

01 Record summary statistics for **mean_RT_ms**, **length**, **log_frequency**, **AoA**, and **OLD20**

02 What does the RT distribution look like?

03 Create a quick scatterplot of **mean_RT_ms** against **log_frequency**

04 Are there obvious outliers or implausible values?

05 Is accuracy very low, very high, or roughly acceptable?

06 Are there missing values?

07 Does anything make you worry about interpreting RT at face value?

CORE MODEL

# Do these predictors explain unique variance in RT?

**mean_RT_ms ~ length + log_frequency + AoA + OLD20**

- **Dependent variable**: mean_RT_ms
- **Predictors**: length, log_frequency, AoA, OLD20
- Ask two things

Does the overall model explain RT variation?

Which predictors show unique associations once the others are included?

Do not add extra predictors unless the core model has been completed and interpreted clearly

INTERPRETATION

# Turning coefficients into claims

For each predictor, note

coefficient sign

confidence interval or p value

plain-language interpretation

Good question

Do frequency and AoA both matter once the other predictors are included?

Better question than “Which is strongest?”

Which predictors show the clearest unique association in the model?

If comparing relative strength, use **standardised estimates** only

Clarify that “independent contribution” here means a unique association in the model, not proof of a separate causal mechanism

OPTIONAL EXTENSION

# Can you apply the same logic to a new dataset?

- If time permits, repeat the workflow on a comparable **Chinese dataset**
- Make predictions before running the model
- Identify which predictors are close equivalents of the English analysis
- Ask
- Which patterns look similar?
- Which look different?
- Are any differences linguistic, orthographic, or methodological?

CONCEPTUAL PAYOFF

# What have we learned?

Lexical decision RT is not random

It is systematically shaped by lexical properties

Regression helps us ask whether predictors make **independent** contributions

These predictors tell us something about the structure of the lexicon

Lecture 2 link

lexical integration asks how new words come to participate in this same structured system and competition